

FINAL YEAR PROJECT REPORT

Spatiotemporal Analysis of Urban Heat Island Effect in Multan, Pakistan (2013–2023)

Using Google Earth Engine, Machine Learning, Landsat 8 C2 L2 Imagery,
and Remote Sensing Spectral Indices

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1. Introduction

Urban Heat Island (UHI) effect refers to the phenomenon where urban areas experience significantly higher temperatures than surrounding rural regions due to human activities, land use changes, and reduced vegetation cover. Multan, one of Pakistan's hottest and most rapidly urbanizing cities, presents a critical case study for understanding spatiotemporal UHI dynamics.

This study leverages Google Earth Engine (GEE) to analyze Landsat 8 Collection 2 Level 2 satellite imagery across seven years (2013, 2015, 2017, 2018, 2019, 2021, and 2023), focusing on the summer season (May–September) when UHI effects are most pronounced.

1.1 Study Area

The study is centered on Multan city (30.1575°N, 71.4753°E) with a 25 km radius Area of Interest (AOI) encompassing approximately 1,963 km². This buffer captures urban, peri-urban, agricultural, riparian, and barren land cover types.

1.2 Objectives

- Analyze Land Surface Temperature (LST) trends across Multan from 2013 to 2023
- Compute spectral indices: NDVI, NDBI, and MNDWI for land cover characterization
- Classify land surface zones using unsupervised K-Means clustering
- Assess spatial vulnerability to heat stress across different land cover types
- Quantify the LST–NDVI relationship to understand vegetation cooling effects

2. Methodology

2.1 Data Source

Landsat 8 OLI/TIRS Collection 2 Level 2 (LANDSAT/LC08/C02/T1_L2) imagery was retrieved via Google Earth Engine. Images were filtered to the study period (May 1 – September 30 per year) with a cloud cover threshold of less than 10%. Cloud masking was applied using the QA_PIXEL and QA_RADSAT quality bands, excluding fill, dilated cloud, cloud shadow, cloud, and cirrus pixels.

2.2 Spectral Indices

The following spectral indices were derived for each year:

- Land Surface Temperature (LST): Derived from ST_B10 thermal band, scaled to °C
- NDVI (Normalized Difference Vegetation Index): $(SR_B5 - SR_B4) / (SR_B5 + SR_B4)$
- NDBI (Normalized Difference Built-up Index): $(SR_B6 - SR_B5) / (SR_B6 + SR_B5)$
- MNDWI (Modified Normalized Difference Water Index): $(SR_B3 - SR_B6) / (SR_B3 + SR_B6)$

2.3 Zonal Classification

An unsupervised K-Means clustering algorithm (k=4) was applied to the 2023 composite image stack (LST, NDVI, NDBI, MNDWI) using 5,000 training samples at 30 m resolution. Clusters were semantically labeled as Water, Vegetation, Barren, and Urban based on mean index profiles.

2.4 Vulnerability Assessment

A composite vulnerability index was constructed combining high LST (> threshold), low NDVI, and high NDBI conditions. Pixels were classified into Low, Moderate, and Extreme vulnerability categories and their spatial extents were computed in km².

3. Results

3.1 Temporal Trends in Urban Indices (2013–2023)

The table below summarizes mean spectral indices and LST computed across the 25 km AOI for each analysis year.

Year	LST Mean (°C)	NDVI Mean	NDBI Mean	MNDWI Mean
2013	40.06	0.342	-0.133	-0.265
2015	41.79	0.353	-0.126	-0.284
2017	41.73	0.397	-0.157	-0.287
2018	41.28	0.314	-0.112	-0.252
2019	41.28	0.352	-0.128	-0.280
2021	39.60	0.370	-0.155	-0.261
2023	40.67	0.370	-0.133	-0.287

Key observations: LST peaked in 2015 (41.79°C) and 2017 (41.73°C), coinciding with reduced NDVI values and elevated NDBI. A notable cooling is observed in 2021 (39.60°C), potentially linked to increased vegetation or climatic variability. NDVI showed a modest upward trend from 0.342 in 2013 to 0.370 in 2023, while MNDWI remained consistently negative, reflecting low surface water presence.

3.2 LST Change 2013–2023

Pixel-wise LST differencing (2023 minus 2013) reveals spatially heterogeneous warming and cooling patterns. The urban core exhibits concentrated warming (red pixels), while the Chenab River corridor and vegetated periphery show cooling (blue pixels). Mixed warming signals in peri-urban zones reflect ongoing land conversion.

3.3 NDVI–LST Regression

Linear regression was performed on 1,000 randomly sampled pixel pairs (NDVI > 0) for each year. The 2023 scatter plot confirms a strong negative relationship: as NDVI increases from 0.1 to 0.7, LST decreases from approximately 52°C to 35°C. This confirms that vegetation plays a significant cooling role in the Multan urban landscape.

3.4 Land Cover Zone Classification (2023)

K-Means clustering identified four distinct land surface zones. Statistics per zone are presented below:

Zone	Label	LST (°C)	NDVI	NDBI	Area %
1	Water	36.18	0.055	-0.221	3.4%
2	Vegetation	37.77	0.563	-0.254	30.8%
3	Barren	40.03	0.365	-0.119	37.4%
4	Urban	45.18	0.207	-0.010	28.4%

Urban zones recorded the highest mean LST (45.18°C), nearly 9°C above water bodies (36.18°C). Vegetation zones maintained relatively low temperatures (37.77°C) despite occupying 30.8% of the AOI. Barren land, covering the largest area (37.4%), had intermediate LST of 40.03°C.

3.5 Vulnerability Assessment

The spatial vulnerability analysis classified the AOI into three heat stress categories:

Vulnerability Class	Area (km²)	Coverage (%)
Low	739.5	38.2%
Moderate	1107.3	57.1%
Extreme	91.1	4.7%

Approximately 57.1% of the study area falls under Moderate vulnerability, with 4.7% classified as Extreme — corresponding largely to the dense urban core of Multan and high-albedo barren surfaces. Only 38.2% shows Low vulnerability, predominantly in vegetated and riparian zones.

4. Discussion

The results demonstrate a clear and persistent UHI effect across Multan over the 2013–2023 period. Urban surfaces consistently register 7–9°C higher than water bodies, consistent with UHI

literature for arid South Asian cities. The negative NDVI–LST correlation corroborates the established role of vegetation as a thermal regulator through evapotranspiration and shading.

The absence of a strong monotonic upward LST trend (with 2021 being cooler) suggests that year-to-year variability — likely linked to monsoon intensity, cloud cover patterns, and agricultural cycles — moderates the urban warming signal. Despite this, the persistence of high-LST urban zones and expansion of built-up surfaces (indicated by NDBI trends) signals ongoing vulnerability.

The K-Means clustering effectively delineated ecologically and thermally distinct zones, demonstrating the power of unsupervised learning for land cover characterization in resource-constrained settings. The vulnerability map provides a spatially explicit tool for urban planners to prioritize greening interventions, cool surface materials, and heat emergency preparedness in Extreme and Moderate vulnerability zones.

5. Conclusion

This Final Year Project successfully implemented a cloud-based spatiotemporal analysis pipeline on Google Earth Engine to characterize the Urban Heat Island effect in Multan, Pakistan. Key findings include:

- Mean LST across the AOI ranged from 39.6°C (2021) to 41.8°C (2015), with urban zones consistently 7–9°C hotter than water bodies
- NDVI increased marginally from 0.342 (2013) to 0.370 (2023), while MNDWI remained negative, reflecting limited green and water cover
- LST–NDVI regression confirmed a strong negative relationship, validating the vegetation cooling hypothesis
- K-Means clustering identified Urban (28.4%), Vegetation (30.8%), Barren (37.4%), and Water (3.4%) zones with distinct thermal signatures
- 57.1% of the AOI is under Moderate heat vulnerability; 4.7% faces Extreme heat risk concentrated in the urban core

These findings underscore the urgent need for evidence-based urban heat mitigation strategies including green infrastructure, cool roofing, and riparian zone conservation in Multan's rapidly expanding urban fabric.

6. Tools & Technologies

- Google Earth Engine (GEE) — cloud-based geospatial analysis platform
- Landsat 8 OLI/TIRS — Collection 2 Level 2 surface reflectance and temperature
- Python (Google Colab) — geemap, pandas, numpy, matplotlib, scipy
- K-Means Clustering — `ee.Clusterer.wekaKMeans` (k=4)
- Spectral Indices — LST, NDVI, NDBI, MNDWI computed at 30 m spatial resolution